



Homeostatic feelings and the biology of consciousness

In a new theory of consciousness, Antonio and Hanna Damasio argue that interoception and the generation of homeostatic feelings are crucial to understanding how conscious states emerge.

Consciousness occurs when mind contents, such as perceptions and thoughts, are ‘spontaneously identified as belonging to a specific organism/owner’. The identification is provided by a continuous flow of ‘homeostatic feelings’, which translate the process of life regulation and include salient fluctuations, e.g. hunger, pain, wellbeing, and states closer to equilibrium, e.g. plain feelings of life/existence.

We propose that homeostatic feelings were the inaugural phenomena of consciousness in biological evolution and were selected because the spontaneous information they provided regarding the current state of life regulation conferred extraordinary advantages to the respective organisms. The ‘knowledge’ carried by conscious homeostatic feelings enabled overt guidance of life regulation.

We also outline a mechanism for generating homeostatic feelings based on a two-way interaction between (i) the nervous system, specifically neural elements of early interoceptive systems; and (ii) non-neural components of the organism, the body’s interior, namely, viscera and circulating chemical molecules. Feelings emerge from this interaction, as hybrid and continuous phenomena, related simultaneously to neural/representational phenomena, and to non-neural/bodily processes. They identify the organism in which they inhere by continuously generating the ‘feelingness’ of traditional philosophical definitions of consciousness.

The problem of consciousness

Neuroscience has been remarkably successful at elucidating the physiological mechanisms behind some of the most complex functions of living organisms, namely memory, emotions, language, attention, and intelligence. When it comes to consciousness, however, the situation could not be more different. Although a variety of solutions have been proposed, none has emerged as satisfactory. Strangely, the failure has been accompanied by the notion that the problem cannot be solved, at least within the realm of biology,¹ and that the solutions, if they exist, might actually be found in the domain of physics.² The multiplicity of conceptions associated with the term consciousness looms large among the difficulties faced by these studies. It is often not clear which specific conception is targeted by the proposed solutions. In search of mechanisms responsible for subjectivity, the prominent solutions often focus,

instead, on mechanisms underlying other complex processes, such as attention and integration of information.^{3–5} That is why we begin by attempting to make clear what we regard as the essence of consciousness.

Consciousness

Consciousness occurs when the contents of mind—what we perceive and recollect and ponder—are ‘spontaneously identified as belonging to a specific owner’: our own organism, located in its body. This definition applies to any ‘scale’ of mind—large and overflowing with contents, as when we contemplate a complex problem, or small, with only a few contents, as when we wake up from deep sleep. Consciousness makes its appearance when a mind process—of large scale or small—becomes enriched by ‘knowledge’ that links it to a specific organism. Consciousness should not be confused with the mere integration of images, or with the scale of that integration, or with the attention accorded to some images in detriment of others. ‘We become conscious when we know, without any question being asked, that the contents of our minds belong to our respective bodies.’

Sensing and minding

The phenomenon of consciousness needs to be distinguished from the simpler ability to ‘sense’ or ‘detect’ objects and conditions in the environment.⁶ Numerous living species, from bacteria to plants, are capable of sensing/detecting and even responding to what is sensed. It is likely that sensing, whose physiology hinges on cell membranes, provides a deep biological foundation for more complex processes of ‘minding’ and ‘consciousness proper’ that subsequently appeared in evolution in more complex organisms. But sensing does not require the sensing organism to possess a nervous system, nor does the sensing organism make internal representations of what is sensed or detected via generating ‘mapped patterns’ or ‘images’. Sensing dispenses with representations.

By contrast, we see consciousness as involving and depending on representations of the kind that constitute minds. The most obvious and common kind of representation takes the form of a ‘mapped pattern’ or ‘image’ of objects, events and qualities (Fig. 1).



Figure 1 Photographer Martin Liebscher plays the piano and watches himself play, perhaps the ultimate stage of consciousness! (C-print diasec, Collection of A. and H. Damasio).

Minds are made of flowing streams of sensory representations or images constructed with the help of a nervous system.^{6,7} In essence, the images are of two kinds. The usually dominant kind of image ‘represents the world surrounding each organism’, a world which includes other living organisms, varied objects and physical structures, and their interactions. Those images are based on exteroceptive sensory processes (e.g. vision, hearing, touch, smell and taste) and map an extraordinary repertoire of the universe external to the organism: perceptions (actual or recalled from memory), thoughts we hold relative to some item we perceive or retrieve from memory, the verbal and syntactical translation of imagetic contents, as well as the conclusions we reach when we reason about a specific issue by logically manipulating images.

The other kind of images is usually less dominant and originates in the organism’s interior via interoception. Those images represent aspects of viscera and visceral states—a racing heartbeat, the flow of air in the respiratory system, a gut colic, the contraction or dilation of blood vessels in the thick of the skin—and include the intensity and quality, pleasant or unpleasant, of what is perceived, e.g. the more or less intense burning pain caused by acid indigestion, or the degree of wellbeing caused by extensive relaxation of smooth and striated muscle fibres throughout the body. Such images manifest themselves as the class of feelings generated by life regulation. Those feelings naturally reflect an organism’s management of existence and are known as homeostatic, to distinguish them from the emotional feelings caused by the triggering of emotions. As we point out, the physiological apparatus that supports the making of feelings is radically different from that of exteroception. (For the sake of brevity, we leave out another perceptual channel, proprioception, which is primarily concerned with representing musculoskeletal structures and the movements they execute but also contributes to homeostatic feelings.)

How is consciousness made: a new hypothesis

We propose that the critical mechanism behind consciousness—the identification of a particular mind with a particular organism—is

naturally provided by the continuous flow of homeostatic feelings within that organism. Prominent examples of salient homeostatic feelings include hunger, thirst, pain, wellbeing, malaise, and desire, as well as subtler feelings of existence that follow unimpeded regulatory states closer to homeostatic balance. Such feelings are generated by the ongoing life regulation as it attempts to maintain operations in the homeostatic range. In brief, homeostatic feelings as dramatic as ‘pain’ or as subtle as a feeling of ‘existence’, are generated continuously in the awake state and support the respective continuity of consciousness, provided they operate above a requisite threshold.⁷

How and why do feelings provide the critical knowledge that constitutes consciousness? The answer is transparent: the most distinctive aspect of the feeling process comes from the fact that ‘each homeostatic feeling is itself spontaneously and automatically conscious’. When you feel pain or hunger or wellbeing, you are necessarily conscious of each of these particular states. Your mind is being offered important and conscious information about how life is progressing inside your organism and about what the organism needs. On the basis of that qualitative and graded ‘knowledge’ you can respond accordingly, by making corrections that may help life continue, e.g. pain signals the possibility of tissue damage while hunger signals the need for additional energy sources; wellbeing serves notice that the organism does not have a specific need at the moment and can engage in exploration. Had feelings not been spontaneously conscious they would not have been able to assist living creatures with curating the life process.

We venture to propose that homeostatic feelings were the inaugural phenomena of consciousness and that they were selected and prevailed in biological evolution because the spontaneous knowledge they provided concerning the current state of life conferred an extraordinary advantage to the organisms that were so endowed. The information/knowledge carried by conscious homeostatic feelings enabled organisms to guide life regulation in an overt manner and behave according to their principal interest: survival. Homeostatic feelings made deliberate life regulation possible by going beyond the already existing covert guidance, which

Table 1 Contrasting the anatomical and physiological characteristics of interoception with those that support exteroception, movement and cognition

	Interoception	Exteroception, movement, cognition
Predominant nerve type of axons	Poorly myelinated or unmyelinated (Aδ/C)	Well myelinated (Aα/Aβ)
Mode of signal transmission	Both nonsynaptic and synaptic	Predominantly synaptic
Time scale	Both slow (s/min/h) and fast	Very fast (μs–ms)
Neural processes supported	Interoception/visceroception, affect (moods, emotions, feelings)	Fine perception, learning, reasoning, calculation, language, movement
Blood–brain barrier	Absent or with major gaps	Continuous
Main neurotransmitters and neuromodulators	Monoamine (dopamine, noradrenaline, serotonin), acetylcholine, neuropeptides	Glutamate, GABA

Impaired consciousness

Disorders of consciousness, a common occurrence in clinical neurology, provide an opportunity to interpret the impairment of consciousness in our perspective.

Consciousness can be radically lost in cases of coma caused by focal damage of the nervous system.^{7,9,10} The critical site for lesion-induced coma is the posterior sector of the brainstem region, where structures such as the nucleus tractus solitarius, the parabrachial nucleus, and the peri-aqueductal grey collect signals from the body’s interior. The first integration of feeling-related signals corresponding to large sectors of the body is accomplished in this region. The consequence of its damage is a direct compromise of integrated body representations and a severing of their links to the exteroceptive mental contents; the latter are largely assembled in the posterior cerebral cortices and integrated in cortical association areas. While the cerebral cortex is the base of operations for creating the predominant imagetic contents of minds, the subcortical regions are needed to support the feelings that identify those imagetic contents as belonging to a specific body. Because of the disparity between the way the CNS handles interoception and exteroception, small brainstem lesions can have a devastating effect on consciousness, whereas even extensive lesions of the cerebral cortex may not. Once the integration of interoceptive information, whose functional culmination in the CNS occurs at brainstem level, is no longer possible, the world of exteroception—whose wide base is in the temporal, parietal and occipital cortices—can no longer be grounded on the body to which it belongs.

On the matter of impaired consciousness, we must briefly comment on general anaesthesia, which is by far the most frequent cause of lost consciousness and is both benign and controlled. The anaesthetics that are commonly used in surgery suspend consciousness rapidly, drastically, and temporarily. Their physiological point of action is the plain ‘sensing and detecting’ process, located well below the functional level of minding at which feelings are generated. As noted above, all living organisms including bacteria, share a basic ability to sense/detect and respond. This ability should not be mistaken for the more complex phenomena of feeling and consciousness, which require nervous systems and representations. It is probable, nonetheless, that sensing/detecting/responding, which depend on the operation of cell membranes, are the foundational level of a functional hierarchical chain that eventually enables feeling and consciousness once nervous systems are present. General anaesthetics seem to pre-empt the first link in the chain and thus preclude interoception and exteroception simultaneously. Once feelings are no longer possible consciousness is also suspended. Curiously, anaesthetics also interrupt regular life activities in

plants.^{7,11} At first glance, this is astonishing, since plants do not have nervous systems or feelings or consciousness. However, because general anaesthetics can preclude sensing/detecting processes, the animation of plants and the complex transactions executed by their roots and leaves, also come to a halt. Plants become just as immobile as anaesthetized patients in the operating room, their animation temporally suspended.

Conclusion

In conclusion, consciousness can be found in many complex living organisms though not in all. It cannot be found in inanimate objects, regardless of how complex they may be. Consciousness is present in living organisms capable of constructing sensory representations of components and states of their own bodies, but not in organisms limited to sensing/detecting. Because nervous systems are a requirement for the building of mapped imagetic representations, it follows that only organisms with nervous systems are likely to be conscious. Still, we caution against concluding that because nervous systems are necessary for consciousness they would be solely responsible for consciousness, as is commonly assumed. We believe that consciousness ‘requires a partnership of nervous systems with the bodies they serve’. Most declarations of failure relative to the problem of consciousness, claim that ‘neuroscience’ failed to solve the problem but there is no reason why neuroscience should have solved it alone, because consciousness does not depend on the nervous system alone.

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Competing interests

The authors report no competing interests.

References

1. Chalmers D. Facing up to the problem of consciousness. *J Conscious Stud.* 1995;2:200–219.
2. Goff P. *Galileo's error: foundations for a new science of consciousness.* Pantheon; 2019.
3. Dehaene S. *Consciousness and the brain.* Viking Press; 2014.
4. Tononi G, Boly M, Massimini M, Koch C. Integrated information theory: from consciousness to its physical substrate. *Nat Rev Neurosci.* 2016;17:450–461.
5. Graziano MSA. Understanding consciousness. *Brain.* 2021;144: 1281–1283.
6. Damasio A. *The strange order of things: life, feeling and the making of cultures.* Pantheon; 2018.
7. Damasio A. *Feeling and knowing: making minds conscious.* Knopf/ Pantheon; 2021.
8. Carvalho G, Damasio A. Interoception and the origin of feelings: a new synthesis. *Bioessays.* 2021;43:e2000261.
9. Parvizi J, Damasio AR. Consciousness and the brainstem. *Cognition.* 2001;79:135–160.
10. Parvizi J, Damasio AR. Neuroanatomical correlates of brainstem coma. *Brain.* 2003;126:1524–1536.
11. Baluška F, Yokawa K, Mancuso S, Baverstock K. Understanding of anesthesia—why consciousness is essential for life and not based on genes. *Commun Integr Biol.* 2016;9:e1238118.